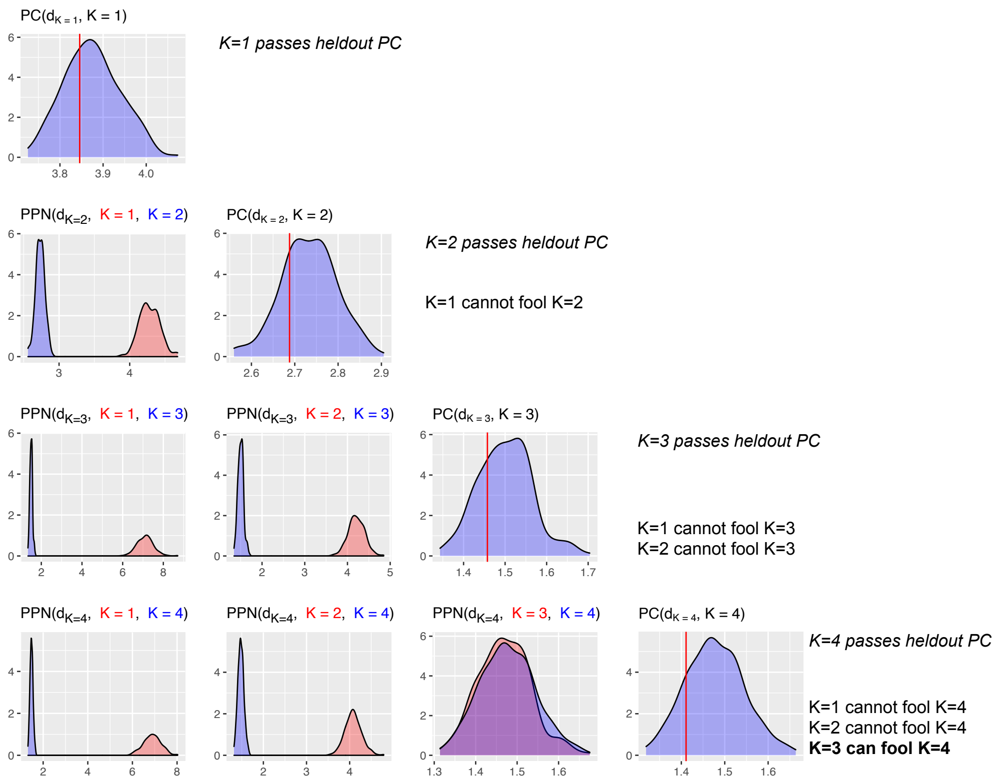

Figure 2: A PPN study of mixture models which suggests $K = 3$ is consistent with the data (no further mixture components are needed). The data are from Figure 1 (left); the true value of K is 3. Along the diagonal are heldout predictive checks; every value of K passes the check. To the left of the diagonal are PPNs, each one checking if a simpler model can fool the model under study. While $K = 1$ and $K = 2$ pass their checks, the PPN shows that they cannot fool $K = 3$, which also passes. On the other hand, $K = 3$ can fool the check for $K = 4$.

can be used to select the number of components in a mixture model. We emphasize that we do not envision the PPN study as a replacement for model selection. Rather, as we discussed, a PPN study can be used as a companion to model selection methods and help to understand the relationships within a collection of "selected" models. In this way, we echo the perspective of Gelman et al. (2020), who contend that presenting multiple models, as opposed to selecting or averaging models, provides a useful picture of the uncertainty inherent in the process of analyzing data. This viewpoint also connects to the "Rashomon effect" as coined by Breiman (2001): there are often many models which have equally good performance. Semenova et al. (2019) expand on this phenomena and define the "Rashomon set" as the set of almost equally accurate models for a given problem. A PPN study determines which models give the same posterior predictive distributions, providing a Bayesian perspective on the Rashomon effect.

In Section 2 we review Bayesian predictive checks, define the posterior predictive null check, and discuss how to use and interpret it. In Section 3 we study and demonstrate the PPN study in different modeling scenarios. With mixtures, we demonstrate how a PPN study can help select the number of components. With probabilistic factor models, we demonstrate how a PPN study can help understand relationships between different classes of models, such as linear models and models based on neural networks. Finally, we analyze a dataset from the literature on predictive checks to show how a PPN study can enhance the practice of Bayesian model criticism.

2 Posterior predictive null checks

2.1 Bayesian model criticism with posterior predictive checks

We want to analyze a dataset $\mathbf{x}_{\text{obs}}$ with Bayesian model A. The model has latent variables θ and is defined by its joint,

$$\mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\text{obs}}, \theta_{\mathrm{A}}) = \mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\text{obs}} \mid \theta_{\mathrm{A}}) \mathrm{p}_{\mathrm{A}}(\theta_{\mathrm{A}}). \tag{1}$$

Bayesian analysis proceeds by evaluating the posterior

$$\mathrm{p}_{\mathrm{A}}(\theta_{\mathrm{A}} \mid \mathbf{x}_{\text{obs}}) = \frac{\mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\text{obs}}, \theta_{\mathrm{A}})}{\int \mathrm{p}_{\mathrm{A}}(\mathbf{x}, \theta_{\mathrm{A}})\, \mathrm{d}\mathbf{x}} \tag{2}$$

and the corresponding posterior predictive

$$\mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\text{rep}} \mid \mathbf{x}_{\text{obs}}) = \int \mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\text{rep}} \mid \theta_{\mathrm{A}})\, \mathrm{p}_{\mathrm{A}}(\theta_{\mathrm{A}} \mid \mathbf{x}_{\text{obs}})\, \mathrm{d}\theta_{\mathrm{A}}. \tag{3}$$

The posterior distribution of θ helps us investigate the latent variables; the posterior predictive provides a distribution of new data.

Many applications of Bayesian statistics end here. Having defined the model, we use its posterior and posterior predictive to their intended purposes. But this is where the activity of Bayesian model criticism begins. Is the model of Eq. 1 a good model of the data? Does it capture the properties of the data that are important to us? If not, in what ways does it fall short?

One of the foundational methods for Bayesian model criticism is the *posterior predictive check* (PPC), an idea that adapts classical goodness-of-fit testing to Bayesian statistics (Guttman, 1967; Rubin, 1984). The central premise of a PPC is that if a model is good then its posterior predictive distribution will capture the true distribution of the data. If the observed data is plausible under this predictive distribution then the model has "passed" the check. Notice that this idea takes a Bayesian approach to modeling and a frequentist approach to checking.

There are several ingredients in a PPC. The first is the *diagnostic statistic* $d_{\mathrm{A}}(\mathbf{x})$. It is a function of observable data that measures the incompatibility between $\mathbf{x}$ and model A. As discussed in Section 4.3 of Gelman et al. (1996), the choice of diagnostic should capture the aspects of the model we are interested in checking. For example, a diagnostic which assesses the overall fitness of a model is the χ^2 diagnostic:

$$d_{\mathrm{A}}(\mathbf{x}) = \sum_{i=1}^{n} \frac{(x_i - \mathbb{E}_{\mathrm{A}}[x_i|\mathbf{x}_{\mathrm{obs}}])^2}{\mathrm{Var}_{\mathrm{A}}(x_i|\mathbf{x}_{\mathrm{obs}})}, \tag{4}$$

where $\mathbb{E}_{\mathrm{A}}$ and $\mathrm{Var}_{\mathrm{A}}$ are expectation and variance with respect to $\mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep}}|\mathbf{x}_{\mathrm{obs}})$ (i.e. model A). In this paper, we consider such model-dependent diagnostic functions in order to assess whether one model can fool another model. In other contexts, the diagnostic might not explicitly depend on the model.

A second ingredient is the *reference distribution*. When the model is adequate, the reference distribution is the distribution of the diagnostic $d_{\mathrm{A}}(\mathbf{x})$ from which we expect the observed diagnostic was drawn. For their reference distribution, Guttman (1967) and Rubin (1984) use the posterior predictive of the diagnostic $\mathrm{p}_{\mathrm{A}}(d_{\mathrm{A}}(\mathbf{x}) \,|\, \mathbf{x}_{\mathrm{obs}})$, which is derived from Eq. 3. If model A is a good model then the posterior predictive of the diagnostic will capture the distribution of the observed diagnostic.

The goal of a PPC is to evaluate whether the observed diagnostic $d_{\mathrm{A}}(\mathbf{x}_{\mathrm{obs}})$ could have plausibly come from the reference distribution. The final ingredient of a PPC is a *measure of surprise*, a method to assess whether an observed value was drawn from a reference. One common approach is to use a p-value, a tail probability. A posterior predictive p-value is

$$p_{\mathrm{post}} = \mathrm{P}(d_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep}}) \geq d_{\mathrm{A}}(\mathbf{x}_{\mathrm{obs}}) \,|\, \mathbf{x}_{\mathrm{obs}}) \qquad \mathbf{x}_{\mathrm{rep}} \sim \mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep}} \,|\, \mathbf{x}_{\mathrm{obs}}). \tag{5}$$

Here a small p-value indicates a poor model: the observed $d_{\mathrm{A}}(\mathbf{x}_{\mathrm{obs}})$ is too surprising under the posterior predictive. Note that a p-value is just one way to locate $d(\mathbf{x}_{\mathrm{obs}})$ in its reference distribution; graphs and other measures of surprise provide good alternatives (Gelman et al., 1996; Gelman, 2004; Bayarri and Castellanos, 2007).

PPCs are an intuitive method for assessing the quality of a Bayesian model, but their statistical properties have also been criticized. The central issue is that the PPC uses the data twice, once to construct the reference $\mathrm{p}_{\mathrm{A}}(\mathbf{x}_{\mathrm{rep}} \,|\, \mathbf{x}_{\mathrm{obs}})$ and once to provide